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SYSTEMATIC REVIEW

Injury Reduction Programs for Reducing the Incidence of Sport-Related Head and Neck Injuries Including Concussion: A Systematic Review

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Abstract

Background Sport-related head and neck injuries, including concussion, are a growing global public health concern with a need to explore injury risk reduction strategies such as neck exercises.

Objectives To systematically review the literature to investigate: (1) the relationship between neck strength and sport-related head and neck injuries (including sport-related concussion (SRC); and (2) whether neck exercise programs can reduce the incidence of (a) sport-related head and neck injuries; and (b) SRC.

Methods Five databases (Ovid MEDLINE, CINAHL, EMBASE, SPORTDiscus, and Web of Science) and research lists of included studies were searched using a combination of medical subject headings and keywords to locate original studies which reported the association between incidence of head and/or neck injury and neck strength data, or included a neck exercise intervention either in isolation or as part of a more comprehensive exercise program.

Results From an initial search of 593 studies, six were included in this review. A narrative synthesis was performed due to the heterogeneity of the included studies. The results of two observational studies reported that higher neck strength, but not deep neck flexor endurance, is associated with a lower risk of sustaining a SRC. Four intervention studies demonstrated that injury reduction programs that included neck exercises can reduce the incidence of sport-related head and neck injuries including SRC.

Conclusion Consideration should be given towards incorporating neck exercises into injury reduction exercise programs to reduce the incidence of sport-related head and neck injuries, including SRC.

Systematic Review Registration PROSPERO (registration number: 194217).

Extended author information available on the last page of the article

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Infographic

Sports Medicine

Injury reduction programs which include neck exercises may reduce the incidence of sport-related head and neck injuries including concussion: A systematic review

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Objectives: To systematically review the literature to investigate: (1) the relationship between neck strength and sport-related head and neck injuries (including sport-related concussion (SRC); and (2) whether neck exercise programs can reduce the incidence of (a) sport-related head and neck injuries; and (b) SRC.

Methods: Five databases and research lists of included studies were searched using a combination of medical subject headings and keywords to locate original studies which reported the association between incidence of head and/or neck injury and neck strength data, or included a neck exercise intervention either in isolation or as part of a more comprehensive exercise program.

What is already known about this subject?

The incidence and burden of sports-related head and neck injuries is high.

Adolescents and females are most at-risk and take longer to recover from SRC compared with adult males.

What are the new findings?

Athletes with greater neck strength have a lower risk of sustaining a SRC.

Injury reduction programs which contain neck exercises can reduce the incidence of sport-related head and neck injuries including SRC.

Results: From an initial search of 593 studies, six were included in this review. A narrative synthesis was performed due to the heterogeneity of the included studies.

Objective 1: Two observational studies (total n=6792 male and female high school and university athletes) reported that higher neck strength, but not deep neck flexor endurance, is associated with a lower risk of sustaining a SRC.

Objective 2: Four intervention studies (total n=3953 predominantly male adolescent and adult rugby union players) demonstrated that injury reduction exercise programs that included neck exercises can reduce the incidence of sport-related head and neck injuries including SRC.

Conclusion: This review provides evidence that greater neck strength and participation in injury reduction exercise programs, which include neck exercises, can reduce the incidence of sport-related head and neck injuries, in particular SRC. The inclusion as well as types of neck exercises incorporated into training programs for contact sport athletes should be further explored particularly in adolescent and female athletes who appear most likely to benefit.

Key Points

This review provides evidence that higher neck strength is associated with a lower risk of sport-related concussion (SRC).

There is clinical worthwhile evidence to support the inclusion of neck exercises into injury reduction programs to reduce the incidence of sport-related head and neck injuries including SRC.

We recommend that future research focuses on the neuromuscular responsiveness of the neck muscles of athletes, particularly of adolescents and females who appear most likely to benefit from this intervention. Neuromuscular neck training in athletes should enable generation of the maximal amount of tension in the shortest amount of time in order to enhance head-to-neck coupling and thus make athletes better able to resist external forces.

1 Introduction

Head and neck injuries in sport are a growing global public health concern with practical implications for sports participation [1, 2]. The risk of sustaining a sport-related head or neck injury is greater in sports which involve body contact, projectiles, additional equipment (such as hockey stick) and high speeds [3]. The personal and societal costs of severe head and neck injuries can be very high [3]. Although catastrophic head and neck injuries in sport are rare, head and neck injuries are the third leading source of severe injuries sustained by National Collegiate Athletic Association student-athletes across 25 different sports in the United States [4].

In athletes, the most common head and neck injury is sport-related concussion (SRC) [5]. The 5th Concussion Consensus Conference in Berlin (2017) defined SRC as a traumatic brain injury induced by biomechanical forces [6, 7]. From a biomechanical perspective, SRC is the result of a blow to the head or body, where resultant inertial impulse force is transmitted through the neck column to the head. From a pathological perspective, the constellation of physical, cognitive and emotional symptoms of SRC are typically temporary with most individuals recovering symptomatically within 10–14 days [6]. Upon imaging, there is typically no evidence of structural injury, which suggests that SRC should be considered as a functional injury, rather than structural [6]. Clinical examination is used to confirm the diagnosis of SRC based on two primary criteria: (1) mechanism of injury (e.g. direct or indirect impulsive force acting on the head); and (2) the reporting of one or more signs or symptoms of SRC [7].

Several studies have shown significant upward trends in the incidence of SRC in recent years [8–11]. The incidence of SRC varies by sport and the level of participation, with high school athletes having a disproportionally higher number of cases each year relative to other comparable age cohorts [11]. The SRC rate more than doubled over a 7-year period across nine high school sports between 2005 and 2012 [8] and was reported to increase 4.2-fold across 12 high school sports, between 1997 and 2008 [12]. The increase in SRC rates is attributed to a combination of factors including improved awareness and surveillance of SRC, changes in working definition of SRC and a lower diagnostic threshold for SRC [12].

There are an estimated four million diagnosed cases of SRC in the United States (US) each year, with evidence to suggest as many as half of these occur in younger athletes [13], and based on recent trends this rate may still be rising [8, 12]. Along with age [14], sex is also a risk factor for SRC, with females being at greater risk of sustaining a SRC [12, 15]. A recent systematic review reported that females demonstrated significantly higher incidence of SRC when compared with males for both football/soccer (rate ratio = 1.76, 95% CI 1.43–2.16) and basketball (rate ratio = 1.99, 95% CI 1.56–2.54); $p < 0.01$) [16]. Females and younger athletes also tend to have a prolonged recovery compared to adult male athletes. A systematic review and meta-analysis comparing recovery times amongst collegiate and high school athletes demonstrated that collegiate athletes reported symptom resolution at 6 days compared to 15 days for high school athletes [17]. A retrospective analysis of medical records of concussed adolescent athletes identified a significant difference in the number of days until complete symptom resolution between male and female athletes (11 vs. 28 days, $p < 0.001$, respectively) [18].

Likely the greatest known intrinsic risk factor an athlete can have for SRC is a previous history of concussion [19]. Athletes with a history of SRC are at least twice as likely to

1.	Exp head injury	(304026)
2.	Exp neck injury	(14244)
3.	Exp SRC	(12955)
4.	Exp muscle strength	(63370)
5.	Exp neck strength	(15578)
6.	Exp Sport	(52390)
7.	1 or 2 or 3	(319586)
8.	4 or 5	(78486)
9.	7 and 8 and 9	(17)

Fig. 1 Search strategy and results in brackets for EMBASE

suffer future SRC and may be as much as 5–10 times more likely [19–22]. Studies have also identified an association between repeated SRC and greater cognitive impairment, risk of depression and risk of lower extremity injuries [23, 24].

Given the potentially serious consequences associated with sport-related head and neck injuries, the focus has rightfully turned towards investigating primary prevention strategies. This is particularly important for young athletes, as their window of susceptibility for sustaining a head and neck injury is greater. Targeted interventions that reduce the burden of disease in at-risk populations are the best way to ease global public health problems such as sports-related head and neck injuries, including SRC. Rule changes [1, 25], athlete technique amendment [25] and selected protective equipment [25] have been shown to mitigate the risk of injury (including incidence of head and neck injury) associated with athletic participation. Neck muscle training is another potential risk mitigating strategy [26, 27].

Lower neck strength capacity, a feature common to both females and young athletes, is postulated to be a modifiable risk factor for head and neck injuries [28, 29]. Decreased neck strength or neck dysfunction may also predispose athletes with a history of SRC to sustaining further SRC, particularly given that neck pain is a common symptom of SRC [30–32]. Neck muscle training may provide an effective and accessible strategy to complement the existing training regimens of athletes most at-risk that will allow them to meet and withstand the biomechanical demands associated with their sport, thus reducing the risk of head and neck injuries.

The objectives of this analysis were to systematically review the literature to (1) investigate the relationship between neck strength and sport-related head and neck injuries including SRC, and; (2) evaluate whether neck exercise programs can reduce the incidence of (a) sport-related head and neck injuries; and, (b) SRC.

2 Methods

2.1 Data Sources and Search Strategy

This review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [33]. The review protocol was registered on PROSPERO (registration number: 194217). This review identified studies for inclusion through five electronic databases: Ovid MEDLINE, CINAHL, EMBASE, SPORTDiscus, and Web of Science, using a combination of medical subject headings (MeSH) and keywords. Manual searches of the reference lists of the included studies were also performed by two authors (KP and TV). An example of the search strategy for EMBASE can be found in Fig. 1.

2.2 Eligibility Criteria for Selecting Studies for the Systematic Review

Studies were limited to human studies from first record until May 2020 based on the following inclusion criteria:

1. Original data (using observational or experimental research designs);
2. *Participants* study sample derived from a population of athletes, regardless of age or sex;
3. *Intervention* for intervention trials, the intervention arm must include a neck exercise intervention completed either in isolation or as part of a more comprehensive exercise program.
4. *Control* for intervention trials, the control arm should not include neck exercises.
5. *Outcome* observational studies which report the association between incidence of head and/or neck injury and neck strength data, or intervention trials which report comparison data for the incidence of head and/or neck injury between the intervention and control arms.

Exclusion criteria included:

1. Not written in English or English transcript unavailable;
2. Review articles, case studies, abstract-only publications, anecdotal reports, commentaries or letters to the editor;
3. Participants not from an athletic population.

2.3 Methodological Quality Assessment

The assessment of methodological quality was undertaken independently by two authors (IG and KP) using the Downs and Black Quality Assessment Tool [34, 35], which has been shown to be reliable and valid for observational and experimental studies [36]. Discrepancies in scoring were resolved through discussion. This tool measures quality in terms of reporting, external validity, internal validity (bias and confounding) and power, with a maximal quality index of 28. Each element of the 27-item checklist is given a score of 1 or 0, except for item-5, which can be scored 0, 1 or 2. For the purposes of this review, item-28 was scored as 0 or 1 depending on the presence of a power calculation [37]. Due to the heterogeneity in study designs of the included studies, any checklist item that was deemed not applicable was removed from the total number of checklist items for that study only. The scoring for each study was summed and converted into percentages by dividing the total score by the number of applicable checklist items to provide the total quality score for each study [36].

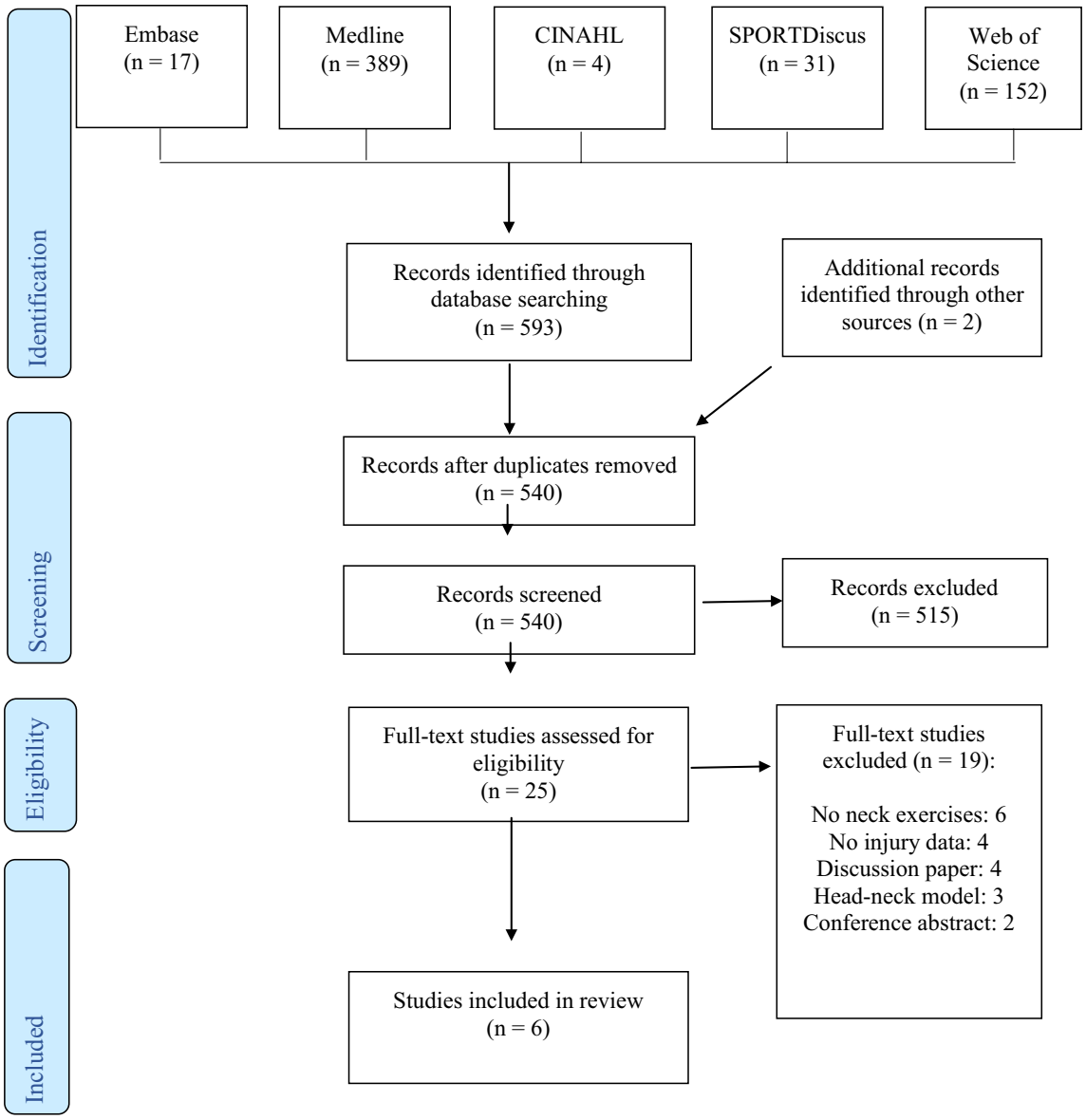

Fig. 2 PRISMA flow diagram of selection process of eligible studies

2.4 Data Extraction

Data were independently extracted by two authors using a pre-tested data extraction form (KP and TV). All authors reviewed the accuracy of the extracted data with any discrepancies resolved through discussion. Data extracted included study design, participant characteristics, intervention and control, results and author conclusions.

2.5 Data Synthesis and Analysis

A meta-analysis was not performed due to the heterogenous characteristics of the included studies in terms of study design (observational [38, 39], cohort [40, 41], cluster RCT [42, 43]), sport, (rugby [41–43] and mixed sports [38–40]), age of athletes (adolescents [38, 40, 42] and adults [39, 41, 43]) as well as differences in injury definitions and intervention characteristics which would have resulted in spurious pooling of injury risk estimates. Data were explored qualitatively for the relationship between neck strength and sport-related head and neck injuries. To investigate whether neck exercise programs can reduce the incidence of sport-related head and neck injuries including SRC, analyses were conducted using STATA (College Station, Texas, USA) epidemiology risk ratio calculator. Risk ratios (RR) were computed using the number of (a) head/neck injuries and (b) SRC incidences per arm in each study and the total number of participants in each arm for all intervention studies

Table 1 Methodological quality and risk of bias assessment

Study	Quality									External validity				Internal validity—bias							Internal validity—confounding						Power	Total Score/ no. of items, %
	1. Reporting (aims)	2. Outcomes	3.Target population	4.Interventions described	5. Confounders described	6. Findings described	7. Estimates of variability	8. Adverse events	9. Missing data	10. Actual probability	11. Representative sample[a]	12. Representative sample[b]	13.Representative process	14. Participant blinding	15. Assessor blinding	16. Retrospective data analysis	17. Follow-up adjustments	18. Statistical tests	19. Intervention adherence	20. Accuracy of measures	21. Variation in population	22. Different groups	23. Randomisation	24. Concealment	25. Confounders adjusted	26. Follow-up	27. Power	
Observational studies																												
Collins et al. [38]	1	1	1	N/A	2	1	1	0	DK	1	1	1	1	N/A	N/A	DK	N/A	1	N/A	1	N/A	0	N/A	N/A	1	0	1	15/20, 75%
Baker et al. [39]	1	1	1	N/A	0	1	1	0	N/A	1	1	1	1	N/A	N/A	DK	0	1	N/A	1	N/A	DK	N/A	N/A	0	N/A	1	12/19, 63%
Experimental studies																												
Attwood et al. [43]	1	1	1	1	2	1	1	0	0	N/A	1	1	1	1	DK	DK	1	1	1	1	1	1	1	DK	1	1	1	22/27, 81%
Hislop et al. [42]	1	1	1	1	2	1	1	0	0	N/A	1	1	1	1	DK	DK	1	1	0	1	1	1	1	1	1	1	1	22/27, 81%
Morrissey et al. [40]	1	1	1	0	0	1	DK	0	0	1	1	1	1	0	0	DK	1	1	DK	1	1	1	0	0	0	DK	1	14/28, 50%
Naish et al. [41]	1	1	1	1	2	1	1	1	N/A	1	1	1	1	N/A	N/A	1	1	1	DK	1	1	1	N/A	N/A	0	DK	0	19/23, 82%

Key: Y = "Yes" = 1/2, *n* = "No" = 0, DK = "Don't know" = 0, N/A "Not Applicable" = item removed from total score of 28

Table 2 Summary of data extraction

Aim 1: To determine the relationship between neck strength and sport-related head and neck injuries (including SRC)

Study (year)	Design	Participants	Aim relevant to review	Measures	Significant Findings	Author conclusions
Baker et al. (2019) [39]	Cohort study	130 university athletes (68 females, 62 males) men's and women's hockey, soccer and basketball. Plus 5 extra male athletes included in post-test analysis only Mean age not stated	Determine if there was a relationship between pre-season deep neck flexor endurance test (DNFET) time of uninjured athletes (UA) compared to injured athletes (IA) SRC was diagnosed using criteria from an inter-professional management team	Deep Neck Flexor Endurance Test (DNFET)	17/135 athletes sustained a SRC (6 females and 11 males). No correlation between pre-season DNFET times and SRC incidence (r = not reported, $p = 0.55$). A moderate correlation between DNFET times and SRC recovery ($r = 0.47$, $p = 0.001$)	Cervical endurance as measured with the DNFET was not predictive of increased risk of sustaining a SRC
Collins et al. (2014) [38]	Cohort study	6662 high school athletes (3002 females, 3660 males) male and female soccer, basketball and lacrosse players. Mean age not specified	Determine if anthropometric measurements captured by athletic trainers can be used to predict SRC risk among high school athletes SRC reported using criteria from the High-School Sports Injury Surveillance study	Composite maximal isometric neck strength value consisting of the average of neck flexion, extension and right and left lateral flexion using a hand-held tension scale	After adjusting for sex and sport, overall composite neck strength remained a significant predictor of SRC $p = 0.004$ The odds of an athlete becoming concussed decreased by 5% for every one-pound increase in composite neck strength (OR = 0.95, 95% CI 0.92–0.98) Athletes who became concussed during the season had 11–22% less overall mean neck strength during pre-season than athletes who did not become concussed	Overall composite neck strength was a significant predictor of SRC among high school soccer, basketball and lacrosse players, indicating higher neck strength is associated with a lower risk of SRC

Table 2 (continued)

Aim 2: To determine whether neck exercise programs can reduce the incidence of a) sport-related head and neck injuries and/or b) SRC

Study (year)	Design	Participants	Aim relevant to review	Intervention and control	Significant findings	Author conclusions
Attwood et al. (2018) [43]	Cluster randomized controlled trial (teams as clusters)	1355 (682 intervention, 673 control) adult male rugby players from 41 clubs. Mean age 25.5 years (SD 5.6)	Evaluate the efficacy of a movement control injury prevention exercise program that included neck strengthening in reducing match injuries that resulted in ≥ 8 days time loss in men's community rugby	Intervention: 42-week exercise program including proprioception, balance and resistance exercises along with isometric neck strengthening of 15 s holds for flexion, extension, right and left side flexion 2–3 times per week Control: dynamic stretching and non-targeted exercises	Intervention group showed a likely beneficial 70% reduction in both burden (RR 0.3; 90% CI 0.2–0.7) and rate reduction ratio (RR) of 0.3 (90% CI 0.2–0.7) for lowering head and neck injury incidence and RR of 0.4 (90% CI 0.2–0.7) for lowering incidence of SRC	The intervention exercise program demonstrated very likely benefit for reducing the risk of head and neck injury and SRC. Higher program adherence was associated with reduced injury burden
Hislop et al. (2017) [42]	Cluster randomized controlled trial (school teams as clusters)	2452 (1325 intervention, 1127 control) adolescent male rugby players from 31 schools. Mean age 16 years (SD 1.15)	Determine the efficacy of a movement control exercise program that included neck strengthening in reducing injuries that resulted in ≥ 24 h time loss in adolescent rugby players and evaluate the dose response to the program on injury measures	Intervention: Resistance and perturbation training delivered by school coaches along with isometric neck strengthening of 15 s holds for flexion, extension right and left side flexion prior to each training session and matches. Delivered between Aug-Dec. 2015 (approx. 20 weeks) Control: dynamic stretching and non-targeted exercises	Intention to treat analysis demonstrated unclear benefit of the intervention program on head and neck injury burden rate ratio (RR) of 0.72 (90%CI 0.51–1.01) and SRC RR of 0.71 (90% CI 0.48–1.05). A dose response was reported: when teams participating in the intervention program completed the program > 3 times per week, the SRC injury burden RR was 0.41 (90%CI 0.17–0.99) compared to the control group	A preventative movement control program that includes neck strengthening likely reduces the incidence of SRC in school aged rugby players. Greater injury reduction is seen with participation in more than 3 sessions per week

Table 2 (continued)

Aim 2: To determine whether neck exercise programs can reduce the incidence of a) sport-related head and neck injuries and/or b) SRC

Study (year)	Design	Participants	Aim relevant to review	Intervention and control	Significant findings	Author conclusions
Morrissey et al.(2019) [40]	Pre-experimental study (no control group—one group pre-post)	119 high school athletes (male American Football, male and female football/soccer and female volleyball). 70 male, 49 females. Mean age not specified	Examine the use of core training as a preventative tool for SRCs in high school athletes SRC diagnosed using the Westmont Hilltop School District criteria	Pre-season 10-week exercise training program which included neck strengthening exercises Control: compared with previous years' data	Expected number of SRCs for study period was 11. Observed number of SRCs was 2. Chi-square contingency test, $\chi 2 = 9.84$ ($p = 0.0017$)	Study showed a statistically significant decrease in SRC rates after participating in pre-season intervention training. This program shows promise as a primary prevention method to reduce SRCs
Naish et al. (2013) [41]	Pre-experimental study (no control group—one group pre-post)	27 male elite rugby union players. Mean age 25.2 years (SD 3.9)	Determine whether 26-week specific neck strengthening intervention program had decreased the number of cervical spine injuries, and the severity (defined in relation to the number of associated missed training or match days) of these injuries Only injuries which resulted in time loss from training or matches were recorded	26-week specific neck strengthening intervention program consisting of two phases; a 13-week pre-season phase and a 13-week maintenance phase Control: compared with previous years' data	Significant ($p = 0.03$) reduction in the number of match-related cervical injuries was evident from 2007–2008 ($n = 11$) to 2008–09 ($n = 2$). No significant differences for the total number of cervical spine injuries ($n = 8$ in 2007–2008, $n = 6$ in 2008–2009) or time loss due to these injuries (100 days in 2007–2008, 40 days in 2008–2009) Non-significant increases in isometric neck strength were found in all directions examined	A significant reduction in the number of match cervical injuries was evident in this neck strengthening intervention study which would be considered as a positive outcome in the professional sporting environment

Key: *DNFET* deep neck flexor endurance test, *IA* injured athletes, *OR* odds ratio, *SD* standard deviation, *CI* confidence interval, *RR* rate ratio, *SRC* sport-related concussion

Table 3 Raw data for each included intervention studies showing total number of participants as well as number of head/neck injuries per arm

Study	Intervention			Control			Head/neck injury RR (95% CI)	Head/neck IRR (95% CI)	Clinical worthwhile effect and certainty of evidence
	Total	Head/ Neck injury Yes	Head/ neck injury No	Total	Head/ neck injury Yes	Head/ neck injury No			
Attwood et al. [43]	682	12	670	673	35	638	0.34 (0.18–0.64) < $p = 0.001$*	0.33 (0.16–0.66) < 0.001*	Yes, certain
Hislop et al. [42]	1325	82	1243	1127	86	1041	0.81 (0.61–1.09) $p = 0.16$	0.82 (0.60–1.13) $p = 0.10$	Yes, uncertain
Naish et al. [41]	27	6	21	27	8	19	0.75 (0.30–1.87) $p = 0.53$	–	Yes, uncertain

Raw data were used to calculate risk ratios (RR) and incidence rate ratios (IRR) with 95% confidence intervals (CI) with associated p values. Certainty of clinical worthwhile effect was based on the point estimate and 95% CI being < 0.90

Key: *Statistically significant; – unable to report; clinically worthwhile = RR point estimate < 0.90; Certainty of evidence = certain (upper and lower CI < 0.90), uncertain (upper CI > 0.91)

Table 4 Raw data for each included intervention studies showing total number of participants as well as SRC per arm

Study	Intervention			Control			SRC RR (95% CI)	SRC IRR (95% CI)	Clinical worthwhile effect and certainty of evidence
	Total	SRC Yes	SRC No	Total	SRC Yes	SRC No			
Attwood et al. [43]	682	12	670	673	33	640	0.36 (0.19–0.69) $p = 0.001$*	0.35 (0.17–0.70) * < 0.001*	Yes, certain
Hislop et al. [42]	1325	51	1274	1127	54	1073	0.80 (0.55–1.17) $p = 0.25$	0.81 (0.54–1.22) $p = 0.15$	Yes, uncertain
Morrissey et al. [40]	119	2	117	505	55	450	0.15 (0.04–0.62) $p = 0.001$*	–	Yes, certain

Raw data were used to calculate risk ratios (RR) and incidence rate ratios (IRR) with 95% confidence intervals (CI) with associated p values. Certainty of clinical worthwhile effect was based on the point estimate and 95% CI being < 0.90

Key: *Statistically significant;– unable to report; clinically worthwhile = RR point estimate < 0.90; certainty of evidence = certain (upper and lower CI < 0.90), uncertain (upper CI > 0.91), *SRC* sport-related concussion

included in this review. In addition, incidence rate ratios (IRR) were calculated using the number of (a) head/neck injuries and (b) SRC incidences per arm in each study and the reported person-time variable (such as total match exposure hours), where available. Results are reported as RR and IRR with 95% confidence interval (95% CI). We also used the smallest clinical worthwhile effect to interpret the clinical significance of the results. The smallest clinical worthwhile effect of an intervention can be defined as the smallest intervention effect that justifies the costs, risks and inconveniences associated with the intervention [44]. The smallest clinical worthwhile effect for each intervention study was calculated using the point estimate for the RR (RR ≤ 0.90, yes, there is a clinical worthwhile effect or RR ≥ 0.91, no, there is not a clinical worthwhile effect) based on the smallest clinical worthwhile difference reported in earlier studies [42, 43]. Clinical worthwhile effects were further quantified as 'certain' favouring the intervention (when both the point estimate and the upper and lower levels of the 95% CI were < 0.90) or 'uncertain' favouring the intervention (when the point estimate was < 0.90 but the upper level of 95% CI was > 0.91). Statistical significance for the α-error was set at $p < 0.05$.

3 Results

After database searches, 593 studies were identified, with 540 studies remaining after duplicates were removed. A total of 25 studies underwent full text screening, of which six were included in the review [38–43]. Results of the study selection are summarised in Fig. 2. Two observational studies (total $n = 6792$ male and female high school and university athletes) explored the relationship between neck strength and SRC [38, 39]. Four experimental studies (total $n = 3953$ predominantly male adolescent and adult athletes)

investigated the effect of an exercise intervention which included neck exercises on incidence of (a) head and/or neck injuries, and/or (b) SRC [40–43].

3.1 Methodological Quality Assessment

Results for observational and experimental studies are presented separately. Table 1 reports the results for methodological quality assessment with quality percentages ranging from 63 to 75% for the two observational studies and 50–82% for the four experimental studies. All studies scored highly for items 1, 2, 3 and 6 related to study quality and 11, 12, 13 related to external validity as well as item 20 (internal validity-accuracy of measures) (Table 1). It was not clear in most of the included studies whether any adverse events were reported or how they handled missing data.

3.2 Data Extraction

Table 2 reports on the data extracted from each study related to study design, participants, aim, outcome measures, intervention and control (as applicable) and results. Raw data for number of SRC or neck injury incidences per study group and total number of participants in each group for the four intervention studies can be found in Tables 3 and 4.

3.3 Relationship Between Neck Strength and Sport-Related Head and Neck Injuries

Two studies examined the relationship between neck strength and SRC using an observational study design with differing results [38, 39]. In the study by Collins et al. [38], high school athletes ($n = 6662$) were recruited from three sports commonly played by male and female participants; lacrosse, football/soccer and basketball. SRC incidence was reported through an existing surveillance system by licensed health practitioners. This study concluded that the odds of an athlete sustaining a concussion decreased by 5% for every one-pound increase in isometric neck strength, with the risk of sustaining a SRC being associated with decreased neck-to-head ratio and decreased neck circumference [38]. These characteristics were more commonly observed in female athletes, with adolescent females reporting a SRC incidence of 4.9 per 1000 athlete exposures (AE) compared to 2.5 per 1000 AE in their adolescent male counterparts [38]. The second study by Baker et al. [39] evaluated a cohort of 130 university-age varsity athletes and reported that sustained deep neck flexor muscle endurance was not associated with SRC incidence (Table 2).

3.4 Effectiveness of Neck Exercise Programs on Reducing the Incidence of Sport-Related Head and/or Neck Injuries

Table 3 reports the RR and IRR (where calculation was possible) for injury reduction programs that included neck exercises for each included intervention study as well as an interpretation of whether there is certain or uncertain evidence of a clinical worthwhile effect. All three studies reported a clinical worthwhile effect, but the certainty of the evidence differed [41–43]. Two cluster RCTs involving a combined total of 3807 rugby union players investigated similar interventions [42, 43]. Hislop et al. [42] recruited adolescent rugby players from independent high schools, while Attwood et al. [43] recruited adult rugby players from community rugby clubs. In both trials, one arm received a pre-activity movement control exercise program (which included neck exercises), while the control arm received dynamic stretching and non-targeted exercises that did not include neck exercises [42, 43]. The RR and IRR calculated for these two studies demonstrated a reduction in head/neck injuries for players completing the intervention (RR = 0.34, IRR = 0.33 [43] and RR = 0.81, IRR = 0.82 [42]). However, the clinical worthwhile effect of the intervention was deemed certain in adults [43] but uncertain in adolescents due to the upper level of the 95% CI being > 0.91 [42] (Table 3). A third pre-experimental study (pre–post-test, no control group) involving 27 professional-level rugby players from a single team explored the effect of a neck strengthening exercise program on neck injuries with uncertain evidence of a clinical worthwhile effect (0.75, 95%CI 0.30–1.87) [41].

3.5 Effectiveness of Neck Exercise Programs on Reducing the Incidence of Sport-Related Concussion

Table 4 reports the evidence from three studies which explored the effectiveness of neck exercises incorporated into a multifaceted injury reduction exercise program on reducing the incidence of SRC [40, 42, 43]. All three studies demonstrated a clinical worthwhile effect with two studies reporting certain evidence. These studies also included the two cluster RCTs by Attwood et al. [43] and Hislop et al. [42]. The RR and IRR calculated for these two studies demonstrated a reduction in SRC for players who completed the intervention (RR = 0.36, IRR = 0.35 [43] and RR = 0.80, IRR = 0.81 [42]). However, the clinical worthwhile effect of the intervention was deemed certain in adults [43] but uncertain in adolescents due to the upper level of the 95% CI being > 0.91 [42] (Table 3). However, for SRC a dose–response relationship was evident for adolescent players in the intervention arm, as players who completed the prescribed program three or more times per

week demonstrated 59% fewer concussions (IRR = 0.41) than those who completed the program less frequently [42]. The third pre-experimental study by Morrissey et al. [40] recruited 119 male and female high school athletes involved in American Football, football/soccer and volleyball. The 10-week pre-season exercise intervention aimed to increase mobility, agility, stability, flexibility, and general strengthening, and included core and neck exercises, with certain evidence of a clinical worthwhile effect (RR = 0.15, 95% CI 0.04–0.62) [40].

4 Discussion

The results of this systematic review provide evidence that higher neck strength may be associated with a lower risk of SRC, with clinically worthwhile evidence to support the inclusion of neck exercises into injury reduction programs for contact sport athletes.

There were two observational studies which examined the relationship between neck strength and SRC [38, 39]. The first study, which included a large sample of high school athletes (total $n = 6662$; $n = 3002$ females, $n = 3660$ males) recruited from three sports: lacrosse, football/soccer and basketball [38]. After adjusting for sex and sport, overall composite neck strength remained a significant predictor of SRC ($p = 0.004$) [38]. While the second study by Baker et al. [39], which included a much smaller sample ($n = 130$) of university athletes, did not demonstrate an association between muscular endurance strength and SRC incidence, the power of their analysis was reduced by a small number of injured athletes ($n = 17$). Despite this, one must not discount the role that neck musculature plays in the dynamic stabilization of the head and neck column. By shifting the focus on neck muscular *strength* to neck muscle *fatigue*, the relationship of the neck muscles to the risk of head and neck injury becomes more plausible. A study by Hendriks et al. [45] reported an increased incidence of SRC during the 4th quarter of rugby games in South African youth rugby, suggesting that fatigue (whole-body and neck) may be a predisposing factor. However, more research is required to elucidate these relationships further.

Three intervention studies investigated the effects of a neck exercise intervention on head and/or neck injuries [41–43]. Despite all three studies demonstrating lower incidences of head/neck injuries in players completing the neck intervention exercise program, only one study demonstrated certain evidence of a clinical worthwhile effect [43]. These three studies all included male rugby players of varying ages and performance levels from adolescent (RR = 0.81, 95% CI 0.61–1.09) [42] to adult community (RR = 0.34, 95% CI 0.18–0.64) [43] through to professional players (RR 0.75, 95% CI 0.30–1.87) [41]. The neck intervention exercise programs were very similar in the two cluster RCTs; however, the length of the intervention period (42 weeks compared with 20 weeks) and the definition of time-loss injuries (> 8 days versus > 24-h) in these two studies differed substantially, which likely had an influence on the certainty of the results (Table 2) [42, 43]. It would seem logical that a longer intervention period would lead to a larger effect. Furthermore, the uncertain evidence of a clinical worthwhile effect in the study by Naish et al. [41] was likely influenced by only a small increase in the players' neck strength. A potential confounding variable is that a ceiling effect may have been present. Given the demands of rugby at the elite level it is likely that these athletes already possessed a level of neck strength capacity that would be significantly greater than the average population, and it therefore would be reasonable to predict that the gains in neck strength would be marginal. It is also possible that the load applied to the exercises in this study was too low given that an earlier study which explored the effectiveness of a neck strengthening exercise program using a similar sample size and playing level of rugby union players demonstrated much larger increases in neck strength over a similar time-frame [46]. This third study also included a much smaller number of players ($n = 27$) compared with the larger studies involving rugby players by Attwood et al. ($n = 1355$) [43] and Hislop et al. ($n = 2452$) [42].

Three intervention studies investigated the effects of a neck exercise intervention on SRC, with two studies demonstrating certain evidence of a clinical worthwhile effect [40, 43], and the third study demonstrating uncertain evidence [42]. The uncertain evidence demonstrated by Hislop et al. [42] is likely related to a dose–response relationship as players who completed the exercise intervention three or more times per week reported 59% fewer SRC than players completing the program less often [42].

The dominant mechanism used to explain why neck strength is protective against head and neck injuries is that stiffer and stronger necks may be better able to stabilise the head during a collision. Neck stiffness is the extent to which the neck resists deformation and this incorporates both active (neck musculature) and passive structures (bone structure, ligament, joint capsule, tendon). Neck strength represents the ability to actively increase neck stiffness through the contraction of neck musculature [47]. Neck strength also requires appropriate neuromuscular control to contract the appropriate muscles quickly enough to help stabilize the head [48, 49]. In a study of professional American Football players, Viano et al. [50] postulated that head displacement, caused by the rapid change in velocity associated with a collision, is the main factor in sustaining SRC and associated brain injuries. Struck athletes were found to have head displacement readings that were proportional to the force in their necks; athletes with stiffer necks had lower

displacement of the head and a reduced risk of SRC [50]. Similar results have also been demonstrated in a controlled laboratory setting, through scaled external perturbations to the head and neck [51]. Tierney et al. [51] reported that greater head accelerations were observed in athletes with decreased neck strength and neck girth compared to athletes with larger, stronger necks. Stronger, stiffer necks are also reported to decrease head accelerations during football/soccer heading [29, 47]. Increased head-to-neck coupling leads to higher effective mass, which can reduce head impact magnitude and subsequent injury risk [47, 52]. However, the relationship between neck strength and head impact magnitude (including linear and rotational acceleration) is conflicting, with significant discrepancies in the literature. Specifically, a cohort study of high school American Football players, reported that linemen with stronger isometric neck strength were 1.75-times more likely to sustain a moderate magnitude head impact, compared to linemen with weaker neck musculature [52]. Similarly, Mihalik et al. [53] did not identify a significant association between increased isometric neck strength capacity and linear and rotational head acceleration in adolescent ice hockey players measured during gameplay via head impact telemetry. In a controlled laboratory setting with external head perturbations, Mansell et al. [54] reported that an 8-week isotonic neck strengthening program did not affect head acceleration during head impacts of male and female National Collegiate Athletics Association, Division I soccer players. All three studies had small sample sizes with 49, 37 and 36 participants, respectively [52–54]. Another possible explanation for these differences is that it may not be neck strength in and of itself which reduces injury risk, but rather the level of neck muscle activation and tension developed before a collision that may reduce linear and rotational acceleration of the head, thus reducing the risk of SRC [12, 27, 55]. The exercise interventions implemented by the studies included in this review may have been successful in reducing head and/or neck injury risk because they improved head-to-neck coupling and neck muscle activation through a multifaceted injury reduction program. Thus, multifaceted exercise interventions which focus on neck and trunk neuromuscular control more broadly may be more effective than isolated neck strengthening exercises. However, this requires further investigation. The threshold of neck strength required to reduce injury risk should also be considered, with the potential for a greater protective effect in athletes with weaker neck musculature such as females and younger athletes.

4.1 Limitations

The protective relationship of greater neck strength in reducing head and neck injury risk, in particular SRC, has been hypothesised for many years, yet the literature to support or refute this claim is equivocal. The investigation of the protective effect of neck strengthening has been limited by the ability to recruit sufficient participants to power an intervention trial. Thus, much of the literature has been based on small sample sizes or observational studies. It is acknowledged that this review includes only a small number of studies of varying study design and quality and the included studies also involved a limited number of sports, limiting the generalisability of the results to sports not included in the research. Similar limitations of heterogeneity of cohort characteristics and small sample sizes have been identified in a recently published systematic review examining the role of neck strengthening and their effect on reducing concussion incidence [56]. This earlier systematic review included three intervention studies, none of which included concussion incidence as an outcome measure. The certainty of the evidence for the inclusion of neck exercises is further limited by the lack of expert consensus and standardization with respect to the design and implementation of neck exercise programs for head and neck injury risk-management to date [56].

Despite our inability to conduct a meta-analysis due to differences in study design, age of athletes, sport, program design and skill level of athletes, we did calculate RR and IRR (where possible) to enable comparison across results for the intervention studies. While this review has demonstrated evidential support for incorporating neck exercises into injury risk reduction programs, the exact type, program frequency and duration are still to be fully explored. Further scientific enquiry is recommended to investigate the type (e.g. isotonic, ballistic isometric, dynamic stabilization) and dose of exercises to maximise injury reduction outcomes relative to sex, age and sporting requirements. Based on available evidence and considering the contextual demands of high-speed collision sports, we recommend that future research focuses on the neuromuscular responsiveness of the neck muscles that would enable them to generate the maximal amount of tension in the shortest amount of time (i.e. rate of force development or contractile impulse) to enhance head-to-neck coupling and thus be better able to resist external forces. We also encourage future research to conduct a priori sample sizes taking into account the incidence rate of sport-related head and/or neck injuries and SRC for each sport as well as considering the smallest clinical worthwhile effect. Further, to assist the interpretation of results, the effect of exposure should be considered with inclusion of person-time variables such as athlete exposure hours or match hours to enable calculation of IRR. Measures which monitor and report athlete adherence to exercise programs are another important consideration, given that the effectiveness of injury reduction programs can be enhanced through higher levels of athlete adherence [57].

5 Conclusion

This review provides limited evidence that greater neck strength and participation in injury reduction exercise programs which include neck exercises may reduce the incidence of sport-related head and neck injuries, including SRC. The inclusion as well as types of neck exercises incorporated into training programs for contact sports should be further explored, particularly in adolescent and female athletes who appear most likely to benefit.

Declarations

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Data availability All data are contained within the manuscript.

Code availability Not applicable.

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